

Augmenting Auditory Attention and Memory to Reduce Cognitive Load in Dual Tasks

A wearable device to augment auditory attention and memory to improve performance in a dual task

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ABSTRACT

We present a wearable device that enhances auditory attention and memory by monitoring the environment for relevant auditory information, storing it, alerting the user to it, and playing back the information. This paper presents the design concept of the wearable device and its empirical evaluation. Using a within-subject design, participants' cognitive and perceptual performance was compared with and without auditory augmentation. Participants were presented with a dual task in which they had to monitor an auditory message for a specific keyword while completing a reading comprehension task. Results showed that the wearable device successfully enhanced auditory attention and memory by taking over the listening task, thereby freeing up cognitive resources for the reading task. In addition, participants' self-assessment indicated that the auditory augmentation was successfully integrated into cognitive and perceptual processes, reducing distraction, and increasing perceived calmness.

CCS CONCEPTS

• **Human-centered computing**; • **Interaction design**; • **Interaction design process and methods**; • **User centered design**; • **Ubiquitous and mobile computing**; • **Ubiquitous and mobile devices**; • **Mobile devices**;

KEYWORDS

Human augmentation, Auditory memory, Wearable device, User-centered design, Divided attention

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1 INTRODUCTION

In daily situations and conversations, we easily miss auditory information because we were distracted, or because of a noisy environment. At the airport we might miss that the departure gate for our flight has been changed, at the hospital we might miss our call to the doctor's office, or we might have problems following conversations because several people are talking simultaneously. In this paper we introduce a wearable device to augment auditory attention and memory by 1) monitoring the environment for relevant auditory information, 2) storing relevant information in the short term, 3) alerting the user to relevant auditory information, and 4) play back this information. As such, the device functions as an augmentation (an extension) of the users' echoic memory, where auditory information are stored for a brief period before being transferred for further processing. In an experimental study, we tested the design concept in a dual task that involved monitoring an auditory message while conducting a reading comprehension task. The main contributions of this paper are threefold: 1) introducing the design concept of a wearable device to augment auditory attention and memory, and experimentally testing the design concept with focus on the 2) augmented perceptual and cognitive performance and the 3) seamlessness of augmentation in terms of perceived distraction.

We provide a short overview of how auditory memory works and where our device supports it, describe the design concept of the wearable device, and present a user study where we tested our design concept.

2 THEORETICAL BACKGROUND

2.1 Augmenting auditory memory

The basic architecture of human memory has been described as a multi-store system by several authors [1, 2] that consists of a sensory store, a short-term store, and a long-term store. The sensory store for auditory information is called echoic store that holds relatively unprocessed information for only about 2 seconds [34]. Recent neurological studies on echoic memory suggest it functions as a real-time monitor to identify new events to which attention may be required [19]. Unattended information is relatively unprocessed, and decays as shown in experiments using shadowing (attending to one auditory channel while ignoring another) [32, 33] or in overload situations when processing capacity is exceeded (e.g., dual task performance) [7, 21]. Information in the echoic store that receive attention are further processed in the subsequent memory systems

such as the Working Memory (WM). According to Baddeley and Hitch [2] the WM consists of a central executive, a visuospatial sketchpad and a phonological loop. The central executive allocates cognitive resources to the two subsystems and controls and regulates cognitive processes. The visuospatial sketchpad stores and processes information in a visual or spatial format while the phonological loop processes spoken and written materials. Thus, any speech-based information (orally or written) is processed by the phonological loop. A central assumption of the model is that available cognitive resources that can be allocated to the two subsystems are limited. When information processing demands exceed available resources, a decline in cognitive performance is observed. The multi-store model described here is a simplified version of how our mind operates and does not depict the full complexity of the memory system. However, the few studies on echoic memory agree that information is stored for only a very brief period, is unprocessed, and decays when not receiving attention [17]. We attempt to augment auditory memory for speech information by extending the limitations of echoic memory and supporting the pre-processing and filtering of auditory information.

2.2 Cognitive limitations in dual task performance

Cognitive limitations often occur in dual task situations where a user performs two tasks simultaneously. Dual tasks compete for cognitive resources and since attentional resources to process information in WM are limited as outlined above [18], the resource competition has a decremental effect on users' performance, also known as dual-task cost [24] [31].

The dual task paradigm is well documented with auditory information in cross-modal settings. Cross-modal attention comes into play when information from different sensory channels (e.g., auditory, and visual) must be processed simultaneously. Scans of brain activity revealed that dual tasks involving written and spoken incongruent sentences activate a similar cortical network even if they involve different modalities (visual and auditory) [22]. Previous studies have shown that background speech impairs reading comprehension [25, 29], proofreading [16], and writing [30]. Furthermore, a meta-analysis of 65 studies on audio-distraction effects found that reading performance is impaired by speech, music, and background noise [35].

3 WEARABLE DEVICE TO AUGMENT AUDITORY MEMORY

We have developed a prototype for a wearable device that supports auditory attention and memory. The conceptual development was based on exploring various usage scenarios in which noise or distractions increase the likelihood to miss important auditory information. For example, announcements in busy and noisy public spaces such as train stations or airports are sometimes difficult to catch. In other situations, we might be distracted (e.g., reading a book or being engaged with a smart phone) and might miss the announcement for the next train station or our turn in the waiting line at the hospital. We imagined that in such situations it would be useful if we could receive an individual notification of information that is important to us and play back the auditory message. With

this in mind, we conceptualized a device that fulfills four important functions to augment auditory attention and memory: 1) monitoring the environment for relevant information, 2) storing relevant information in the short term, 3) providing an alert to relevant information, and 4) play back this information. As shown in Figure 1a), the device uses *binaural microphones* to record environmental sounds. A *voice activity detector* is used to extract speech data from the sampled data of the *sound recorder*, while background noise is suppressed. The voice segment is stored in a transient way in a *voice activity buffer* for a short time (about 5 seconds). Thereby new information is continuously replacing old information in the *activity buffer* so that only the most recent 5 seconds are stored, augmenting (and extending) the users' echoic store for auditory information. Furthermore, the voice segment is sent to a *speech event detector*. The *speech event detector* converts the speech of the voice segment into text using speech recognition. Predefined user-selected keywords (e.g., a specific flight number, train station, or waiting number) are searched for in the generated text data. When a predefined word is found, a *speech notification* is sent to the user at the end of the speech segment (usually at the end of a sentence) in which the keyword was embedded and the auditory message containing the keyword is played back. *Binaural bone conduction headphones* are used for playing back the auditory information allowing the user to still detect sounds from the environment while listening to the replay sound. Figure 1 (b) depicts the workflow of our prototype as applied in the experiment.

As depicted in Figure 2, the device extends the user's echoic memory by buffering auditory information, and directs user's attention to important auditory information so that they can be processed in WM. The automatic detection of relevant information in speech reduces the user's task of monitoring, selecting, and evaluating auditory information, freeing up cognitive resources and reducing WM load.

In recent years various attempts have been made to augment auditory perception. De Godoy et.al. [14], for example, developed a wearable system for early hazard detection for pedestrians. Similarly, Ramadhan et. al [27] presented a wearable intelligent system for visually impaired people that tracks their path and warns them of obstacles. [20] developed a system to support the selective sensing of single sound sources, and [10] provided a visualizer tool that enhances the musical experiences for people with hearing impairments. However, research on wearable systems that enhance auditory attention and memory is still rare.

4 HYPOTHESES

The aim of this study was to verify whether the wearable device frees users' attentional resources and reduces cognitive load by monitoring, storing, alerting the user to relevant auditory information, and play them back. Since the effect of auditory distraction on reading tasks is well documented in the literature [15, 25, 29] and reading and listening to speech requires similar processing resources in the phonological loop [2], we chose a scenario where participants had to conduct a reading comprehension task while monitoring an auditory message for a specific keyword. We propose that the wearable device reduces cognitive load by monitoring and

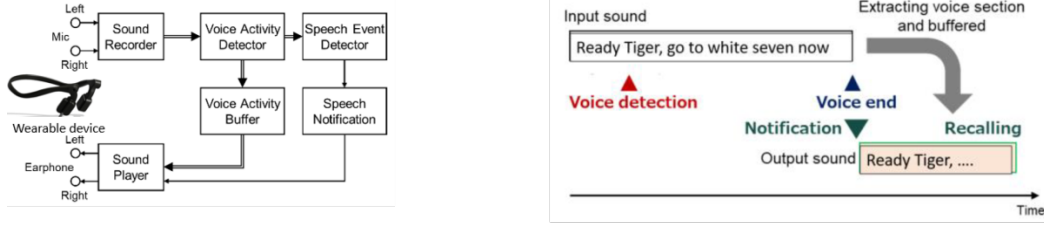


Figure 1: (a) Prototype hearable device system of echoic memory augmentation, (b) Workflow of the prototype.

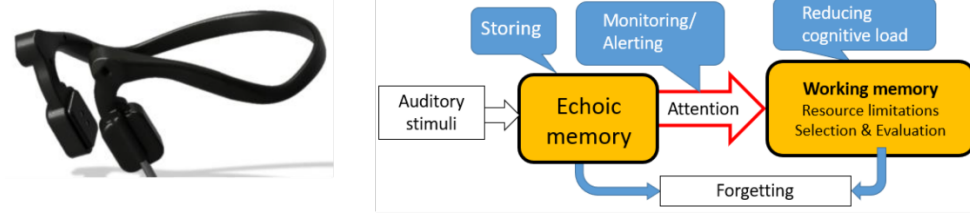


Figure 2: Augmentation of auditory information through a wearable device

evaluating the auditory message and by selecting relevant information (keyword detection) for the user. Consequently, changes in cognitive load should be reflected in reading performance, and we hypothesize that (H1) Reading performance will be higher with augmentation than without. Since the wearable device detects keywords in the auditory message on behalf of the user, we propose that (H2) Participants will detect more keywords with augmentation than without. Additionally we propose that our wearable device augments auditory memory in an unobtrusive manner integrating augmented technology and perceptual and cognitive abilities seamlessly. The wearable device works in the background and only alerts the user and replays messages that are of relevance while on the other hand the user must constantly monitor the auditory message and must be on alert when not using the device. Therefore, we assume that (H3) Perceived distraction is lower with than without augmentation, and that (H4) Perceived calm is higher with than without augmentation.

5 METHOD

5.1 Participants

The required number of participants was calculated with G*Power 3.1.9.4 [12], considering tests for dependent samples with an Effect size $d_z = 0.7$ (medium effect), $\alpha = .05$ and statistic power of 0.80 resulting in a minimum sample of $N = 15$. A total of 16 participants (company employees not involved in the project) took part in the study (age range between 20~65; 2 women, 14 men). The ethical guidelines of the company as well as the ethical guidelines according to the declaration of Helsinki for research involving Human Subjects were maintained. Participation was voluntary and informed consent was obtained from all participants. All participants were of normal hearing.

5.2 Experimental design

A laboratory experiment using a within-factorial design with the two conditions with and without augmentation was conducted. In the augmented condition participants used the wearable device introduced in 3 which automatically detected the keyword in the auditory message, alerted the participant and played it back. The conditions with and without augmentation were counterbalanced. Participants had to attend to two tasks (dual task paradigm) that required language comprehension simultaneously, a visually presented reading comprehension task and a listening task. The two sets of audio messages (listening task) and the two sets of reading tasks were counterbalanced across the study.

5.3 Equipment and setting

A small room with soundproof features that reduced outside noise was used for the experiment. Participants sat in front of a computer screen that displayed the sentences used in the reading task. Participants used arrow keys on the keyboard for selecting the correct word to fill in the blanks in the sentences. Speed of selection and word choice were recorded in a logfile. The auditory message for the listening task was played through a speaker placed on the participant's left side. Detection of the call sign and color and number after the call sign were recorded in a log file. In the augmented condition, the wearable device presented in Figure 1 was used.

5.4 Material

Interactive reading task. Sentences containing a blank were displayed one by one on a computer screen. Participants had to select one out of four words that fit best into the blank (e.g., "Sad" and "happy" are words for _____. Answer options: feelings, opinions, intention, admiration). The task required language comprehension and competed for cognitive resources in the phonological loop with the listening task. The time for reading and answering each sentence was limited to 7 seconds. The time limit was set based

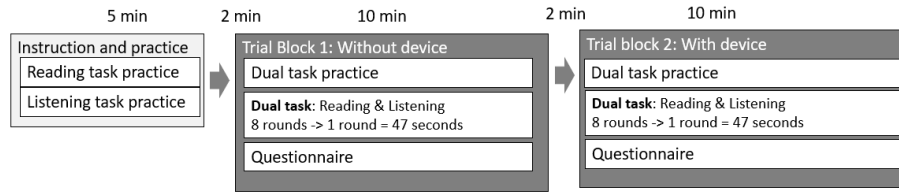


Figure 3: Experiment procedure

on a pilot study with 3 participants at two standard deviations above the average. If the time limit was exceeded the next sentence was displayed, and a *TIMEOUT* was recorded. Participants were instructed to solve as many sentences as possible within one round of 47 seconds. Time taken for each sentence and correct (*TRUE*) and *FALSE* answers were recorded.

Listening task. Sentence samples from the Japanese version of the Coordinated Response Measure corpus (CRM) [3, 13] [3] were used. CRM has been widely used in studies of auditory perception [5, 11, 28]. Using a well-tested set of audio stimuli for the experiment ensured reliability and allowed us to control for other factors that might affect cognitive load in the dual task, such as prior knowledge or task difficulty. The corpus is based on a simple sentence structure following the pattern: “Ready <call sign>, go to <color> <number> now”. A total of 80 sentences were created making sure that all animals, colors, and numbers were balanced. For each round of listening, 12 sentences were played in a row with 0.5 seconds in-between each. Each listening round took 47 seconds. The call sign “Tiger” that had to be detected by participants was randomly placed in sentences 7 till 11. Color and number occurring after the call sign had to be remembered to ensure that participants paid attention to the listening task.

Questionnaire. Perceived distraction was measured according to the study by Moore and Urakami [23] using four items of the Adapted attentional behavior rating scale [26] (internal consistency of Cronbach’s Alpha = 0.89 [23]). Additionally, perceived calm was measured using two items from Moore and Urakami [23] (internal consistency of Cronbach’s Alpha = 0.80). Furthermore, task easiness was assessed with one question. All items were assessed by 7-point Likert scales ranging from 1 (strongly disagree) till 7 (strongly agree).

5.5 Procedure

As displayed in Figure 3, first participants received instructions and practiced the reading and listening task. The experiment was divided into two blocks one with and the other without augmentation (counterbalanced). In each Trial Block, participants conducted eight rounds of the dual task (listening to the audio message and reading task). Participants used the arrow keys on the keyboard for selecting the words for the reading task. At the same time, participants had to press the Q-button on the keyboard as soon as they heard the keyword *Tiger* in the audio message.

After each round participants had to select the correct color (one out of four) and number (one out of eight) that had appeared after the keyword *Tiger* in the auditory message. After each set of eight rounds participants answered the questionnaire about perceived

distraction, calm and task easiness. One session took about 30 minutes.

5.6 Statistical analysis of data

Hypotheses were tested on an alpha-level of 0.05. Data were tested for normal distribution and homogeneity of variances and when assumptions were violated non-parametric tests were used. All effects were tested for order effects between the two Trial Blocks. Reading comprehension performance was only assessed for tasks completed before the keyword *Tiger* appeared in the auditory message since the wearable device assists users in monitoring and recognizing the relevant keyword and therefore an effect on the reading comprehension task was expected before the occurrence of the keyword. TRUE and FALSE rate were calculated by dividing the number of TRUE/FALSE answers through the total number of TRUE+FALSE answers. To normalize (standardize) values for further analysis z-transformation was applied to the TRUE rate and FALSE rate and median corrected z-transformation was used to normalize reaction time values. To account for the trade-off between speed and accuracy a combined performance score was calculated based on Chignell et.al. [8, 9].

$$\text{Combined performance} = -z(\text{Median correct RT}) - z(\text{FALSE rate})$$

6 RESULTS

6.1 Reading task

On average, participants completed a total of 66 ($M = 66.25$, $SD = 12.09$) tasks without augmentation and 68 ($M = 68.19$, $SD = 13.10$) tasks with augmentation. Of these, approximately 60 ($M = 59.81$, $SD = 13.14$) were TRUE and 6 ($M = 6.44$, $SD = 3.50$) were FALSE (including 1-2 TIMEOUTS) without augmentation and 63 ($M = 62.56$, $SD = 13.42$) were TRUE and 5.6 were FALSE (including an average of 1 TIMEOUT) with augmentation.

As shown in Table 1, with augmentation, participants achieved a higher TRUE rate and a lower FALSE rate than without augmentation, TRUE rate: $t(15) = -2.16$, $p = 0.04$; FALSE rate: $t(15) = 2.16$, $p = 0.04$. Differences in reaction time were only marginally significant, $t(15) = 1.46$, $p = 0.16$. However, combined performance (accounting for speed and accuracy by integrating reaction time and FALSE rate) showed that higher performance was achieved with augmentation than without augmentation, $t(15) = -2.74$, $p = 0.01$. No order effects and no differences between reading comprehension sets 1 and 2 were observed for combined performance, $t(15) = -0.61$, $p = 0.55$; $t(15) = -0.17$, $p = 0.86$ respectively.

Table 1: Mean and Standard deviation for reading task, questionnaire, and listening task; * $p < .05$, ** $p < .0$

		Without augmentation M (SD)	With augmentation M (SD)
Reading task	TRUE rate*	0.898 (0.06)	0.914 (0.05)
	FALSE rate*	0.101 (0.06)	0.086 (0.05)
	Average reaction time (RT)	3.317 (0.62)	3.219 (0.66)
	Combined performance*	-0.270 (1.82)	0.196 (1.69)
Questionnaire	Perceived distraction*	5.17 (1.37)	4.64 (1.37)
	Perceived calm**	2.06 (1.34)	2.94 (1.38)
	Task easiness**	2.12 (1.36)	2.93 (1.57)
Listening task	Color + number	14.56 (4.13)	15.31 (1.83)

6.2 Questionnaire

Subjective feedback from participants indicated that they felt less distracted ($t(15) = 2.27, p = 0.04$), calmer ($t(15) = -2.98, p = 0.01$), and found the task easier ($t(15) = -3.57, p = 0.003$) with augmentation than without augmentation (see Table 1 for Mean and Standard deviation). No order effects were observed.

6.3 Listening task

As shown in Table 1, memory for color and number differed only slightly with and without augmentation and no statistical significance was found with a Wilcoxon signed rank test, $z = -1.37, p = 0.08$. There were no order effects (comparing first and second trial) and no stimuli effects for Audio set 1 and 2, $p > .05$ for all comparisons respectively. The number of times participants pressed the Q-button when detecting the keyword *Tiger* did not differ between with ($M = 5.87, SD = 6.92$) and without augmentation ($M = 6.44, SD = 4.80$), $t(15) = 1.06, p = 0.31$. No order effects were observed, $t(15) = -0.81, p = 0.43$. In general, many participants admitted that they forgot to press the button because they focused on remembering the color and number after the keyword.

7 DISCUSSION

The goal of the study was to confirm whether our wearable device can augment auditory attention and memory successfully by reducing the cognitive load for the user in a dual task. The results of the reading comprehension task suggest that the wearable device freed up cognitive resources that were allocated to the reading comprehension task. This was particularly evident in the combined performance score (combination of speed and accuracy), which was higher with augmentation than without, supporting hypothesis H1. The results indicate that participants in the augmented condition transferred the listening task to the wearable device and allocated the freed-up cognitive resources to the reading comprehension task. At the same time, participants' performance on the listening task did not suffer from this transfer, indicating that the wearable device successfully and seamlessly preprocessed auditory perception and filtered auditory information. Without augmentation, participants had to divide their cognitive resources between the reading and listening tasks, which led to a decline in performance - an effect often observed in dual tasks [4, 24, 32]. The results are consistent with the assumption that a pool of resources must be allocated between tasks [18, 24], and that performance is impaired when demand exceeds the availability of these cognitive resources [4, 33].

Accordingly, participants' subjective evaluation of perceived distraction and calmness indicated a similar trend. The wearable device created an interaction that could be described as calm and less distracting allowing participants to focus entirely on the reading task confirming hypotheses H3 and H4. Thus, the wearable device successfully augmented auditory attention and memory to support perception and cognition without being intrusive. The results of our study are consistent with the findings of Moore and Urakami [23] showing that technology perceived as calm reduces distraction and improves performance.

The results in terms of recognition of relevant auditory messages were inconclusive. No differences were found in the number of key words recognized with and without augmentation (H2). Many participants indicated that they had forgotten to press the key when they recognized the keyword. On the other hand, recall of relevant auditory messages was high in both conditions, indicating that participants had not neglected the listening task.

A limitation of the study is the relatively small sample of sixteen participants and the gender imbalance. Studies of new technologies often must make do with a smaller sample for cost and feasibility reasons [6]. The participants in this study worked in the same company in a technology-related field, which may have led to a positive bias toward the use of augmented technology. Another limitation is the rather artificial task using the CRM corpus and a fill-in-the-blank reading task. The use of such standardized tasks increased the reliability of the experiment and allowed us to control for task-related effects on cognitive load. In the future, we plan to test the prototype in real-world scenarios, as described in the usage scenes, with more diverse samples to ensure the validity of our results.

8 CONCLUSION

In this paper, we presented a wearable device that augments auditory attention and memory by 1) monitoring the environment for relevant information, 2) storing relevant information in the short term, 3) alerting the user to relevant auditory information, and 4) playing back this information to the user. The wearable device goes beyond existing augmented technologies that enhance the perception of auditory information by also supporting the cognitive processes for selecting and processing relevant information. In an experimental study we confirmed that the wearable device successfully augmented auditory perception and reduced cognitive load. Subjective evaluation indicated that the augmentation enhanced

auditory perception in an inobtrusive way, seamlessly extending the capabilities of the human body and its senses through technological enhancement. Further studies in the wild with a larger number of users are needed to confirm that the device works just as well in real-world scenarios.

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